ZZ-H-451c
October 7, 1964

SUPERSEDING
Int. Fed. Spec. ZZ-H-00451b (GSA-FSS)
April 5, 1963 and
Fed. Spec. ZZ-H-451a
April 29, 1941

FEDERAL SPECIFICATION

HOSE, FIRE, WOVEN-JACKETED, RUBBER-OR CAMBRIC-LINED, WITH COUPLINGS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers woven-jacketed rubber- or cambric-lined fire hose with couplings.

1.2 Classification.

1.2.1 *Types, classes, and sizes.* Fire hose covered by this specification shall be of the following types, classes, and sizes, as specified (see 6.2):

Type I —Cotton jacket.
Type II —Cotton — polyester jacket.
Type III—Polyester jacket (class B only).
Class A—Double jacket (types I and II only).
Class B—Single jacket.
Size 1½—1½ inch minimum (min) inside diameter (id).
Size 2½—2$^{9}/_{16}$ inch min id.
Size 3½—3½ inch min id (types I and II, class A; and type III, class B only).
Size 4—4 inch min id (types I and II, class B only).
Size 5—5 inch min id (type III only).
Size 6—6 inch min id (type III only)

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

WW-C-621—Coupling Assembly, Hose, (Fire, Woven - Jacketed, Rubber or Cambric-Lined and Unlined Linen).
CCC-T-191—Textile Test Methods.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 601—Rubber: Sampling and Testing.

FSC 4210

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-H-775—Hose, Rubber, or Fabric (Including Tubing); and Fittings, Nozzles and Strainers, Packaging of.

Military Standards:

MIL-STD-129—Marking for Shipment and Storage.

MIL-STD-130—Identification Marking of U. S. Military Property.

MIL-STD-166—Visual Inspection Guide for Rubber Hose.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following publication forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

Underwriters' Laboratories, Inc. (UL) Standard:

UL 19—Standard for Cotton Rubber-Lined Fire Hose.

(Copies may be obtained from the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Illinois 60611; 161 Sixth Avenue, New York 13, New York; or 1655 Scott Boulevard, Santa Clara, California 95050.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies).

3. REQUIREMENTS

3.1 Label of Underwriters' Laboratories, Inc. Hose conforming to UL 19 and to paragraphs 3.3.3.3 through 3.3.3.3.2 (jacket construction and minimum denier of polyester filler yarn); 3.3.6 (weight per length of type III hose); 3.4 (couplings and gaskets); 3.7.6 through 3.7.6.2 (resistance to water); the low temperature and aging characteristics specified in 3.7.2 (ultimate elongation), 3.7.3 (set) 3.7.7 (torsional stiffness ratio and cold tension recovery); and 3.8 (low temperature flexibility) of type III hose; and 3.9 (resistance to fungus), as applicable, will be considered as meeting the requirements of this specification.

3.2 Material. All material shall be new and unused. The hose shall have been manufactured not more than one year prior to date of delivery.

3.2.1 *Rubber.* The rubber or latex used in the lining shall be compounded from natural rubber, synthetic rubber, or a mixture of the two.

3.2.2 *Jacket yarns.* The yarns used in the jacket shall be cotton or polyester yarns as required hereinafter. The yarns shall be well and evenly made and shall be as free from unsightly defects, dirt, knots, lumps, and irregularities of twist as is consistent with the best manufacturing practice.

3.3 Construction. The hose shall consist of a lining with or without rubber backing completely covered with a woven jacket or jackets. The contour at the fold line of the finished hose shall be a smooth continuous curve and shall show no evidence of crease.

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3.3.1 *Lining.* Unless otherwise specified (see 6.2), the lining shall consist of a solid rubber compound. When specified (see 6.2), the lining of types I and II hose may consist of cambric fabric coated with latex or rubber.

3.3.1.1 *Rubber lining.* The rubber lining shall be uniform in thickness and shall be made from a single ply extruded tube or from calendered sheets of not less than three plies, lap jointed and vulcanized in the round into one solid body.

3.3.1.2 *Cambric lining (types I and II).* The cambric shall be rubber coated and shall conform to the following requirements:

Construction	— Warp 105 ends/inch: filler 85 picks/inch.
Thickness	— 0.009 inch.
Weight	— 2.75 ounce/square yard.

3.3.1.3 *Lap.* The lap joints of the calendered lining or of the rubber coated cambric lining shall be as small as is consistent with good manufacturing practice.

3.3.1.4 *Surface.* The waterway surface of the lining shall be practically free from corrugations, pitting, and other imperfections as is consistent with good manufacturing practice.

3.3.1.5 *Thickness of lining.* The thickness of the rubber lining, exclusive of rubber backing, shall be as shown in table I, except that type I or type II hose with a coated cambric lining shall have a tube thickness of not less than 0.015 inch, including inner latex or rubber coat and cambric carrier tested in accordance with 4.3.2.

TABLE I. *Thickness of rubber lining*

Size of hose	Thickness, minimum (inch)
1½ and 2½	0.049
3½, 4, 5, and 6	0.072

3.3.2 *Rubber backing.* The rubber backing, if used, shall be made from calendered sheet. It shall be uniform in thickness and the thickness shall be not more than 0.028 inch measured from the bottom of the impressions of the jacket.

3.3.3 *Jacket.* The jacket(s) shall be seamless and shall have the fillers woven around the lining throughout its length. Class A jackets may be separate or interwoven. At no point shall the liner or the backing show through the outer jacket, nor shall the filler yarn show through the warp yarn. The jacket(s) shall be evenly and firmly loomed and shall be free from knots, lumps, and irregularities in weave.

3.3.3.1 *Type I.* The jacket(s) of type I hose shall be woven from cotton yarns in both the warp and filling directions.

3.3.3.2 *Type II.* The jacket(s) for type II hose shall be woven from cotton yarns in the warp and polyester fiber yarns in the filling directions.

3.3.3.3 *Type III.* The jacket for type III hose shall be made of polyester fiber and shall be seamless and circular woven. The outside diameter of the hose shall be as specified in table II.

TABLE II. *Weight, length, and outside diameter of type III hose*

Size	Weight per length maximum (pounds)	Nominal length (feet)	Outside diameter (inches)	
			Min.	*Max.*
1½	20	50	1.78	1.84
2½	34	50	2.84	2.90
3½	60	50	3.87	3.93
5	15	10	5.38	5.52
6	22	10	6.38	6.52

3.3.3.3.1 *Filler yarn, type III.* The filler yarn shall be continuous-filament polyester fiber having a minimum twist of one turn per inch in the "S" direction. The filler ply yarn or tow shall be a minimum of 6,600 denier for the 1½-inch size, 11,000 denier

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for the 2½-inch size, and 16,500 denier for size 3½, 5, and 6 inches.

3.3.3.3.2 *Warp yarn, type III.* The warp yarn shall be spun from a suitable denier per filament and staple length of first quality polyester staple fiber. The yarn shall be a minimum of 16 plies, except the 1½-inch size shall be a minimum of 9 plies, having a minimum ply twist of 2 turns per inch. The counts of the singles yarn (840 yards per pound) shall be a maximum of 12.

3.3.4 *Inside diameter.* The id of types I, II, and III hose shall be not less than the nominal specified size except that the 2½-inch size shall have an id not less than 2$^{9}/_{16}$-inches.

3.3.5 *Length.* When measured from back to back of the couplings as described in 4.2.2.1, types I and II hose and type III hose sizes 1½, 2½, and 3½ shall be furnished in lengths averaging not less than 25 or 50 feet, as specified (see 6.2), and type III hose of sizes 5 and 6 shall be furnished in nominal lengths of 10 feet (see table II). For type III hose a minus tolerance of 6 inches per 50-foot length, 3 inches per 25-foot length, and 1 inch per 10-foot length will be permitted. Such 50-foot lengths of types I and II hose as have had samples removed for tests, when coupled, shall be accepted as 50-lengths after reattachment of couplings provided no single length is less than 47 feet after the removal of test samples (see 4.1.5). Test samples for 25-foot lengths of types I and II hose and all lengths of type III hose shall be taken from the manufacturer's assembled tubing run and not subtracted from the coupled lengths delivered against the contract.

3.3.6. *Weight, type III.* The maximum weight per length of type III hose without couplings shall be as stated in table II for the corresponding size. Variations in weight proportional to the acceptable tolerances for length will be accepted.

3.4 Couplings and gaskets. Unless otherwise specified (see 6.2), each length of hose shall be fitted with a set, one male and one female, of couplings conforming to WW–C–621. When specified (see 6.2), both couplings shall be either male or female. Unless otherwise specified (see 6.2), couplings shall be type B1 with National Hose (NH) thread form. When specified (see 6.2), couplings for type III sizes 1½ and 2½ hose shall be type D1 with NH thread form. When specified for shipboard use (see 6.2), size 1½ threads shall be the National Hose Coupling (NPSH) thread form with 11½ threads per inch and size 4 threads shall be NH (special) thread form with 6 threads per inch. Each set of B1 couplings shall be provided with three rubber gaskets of rubber meeting the material requirements of the lining of the type of hose with which used. One gasket shall accurately fit the internally threaded swivel of the coupling, and one not less than ⅛ inch in thickness, shall be placed under the end of each expansion ring. When the coupling is made up snugly as in service, the gasket shall be not compressed to such an extent as to project beyond the inside surface of the coupling into the waterway. Each set of D1 couplings shall be provided with one gasket which shall accurately fit the swivel of the coupling. Polyvinyl chloride gaskets may be used in lieu of rubber gaskets.

3.5 Bursting strength. The hose shall not burst at pressures less than those shown in table III for the respective type, class, and size of hose as described in 4.3.2.1. The pressure shall be applied to the hose when it is lying straight and when it is held to a circular radius of 27 inches.

3.6 Hydrostatic properties. The hose shall not leak, sweat, or show any indication of weakness, and there shall be no leaking at

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TABLE III. *Pressure to be applied to hose in burst test*

Type	Class	Size	Pressure p.s.i.
I and II.....	A	**All sizes**	600
I and II.....	B	1½ and 2½	500
I and II.....	B	4	300
III	B	1½, 2½, and 3½	**600**
III	B	5 and 6	**200**

the couplings nor breaking of yarns in the jacket when subjected to the maximum applicable hydrostatic pressures specified in table IV for not less than 5 seconds, and tested as described in 4.3.1 through 4.3.1.6. The pressure shall be applied to the hose when it is lying straight and when it is doubled back on itself to form a sharp kink. The hose when lying straight shall also meet the requirements specified in table IV for elongation, expansion, twists to the right, twists to the left, warp, and rise, tested in accordance with 4.3.1.1 through 4.3.1.6. After passing all tests for elongation, expansion, twist, warp, and rise, the hose shall be branded (see 3.11) to indicate compliance with this requirement.

3.7 Physical properties.

3.7.1 *Tensile strength.* The tensile strength of the lining across the fold shall be not less than 1,200 pounds per square inch (p.s.i.) (see 4.3.2.3.).

3.7.2 *Ultimate elongation.* The ultimate elongation of the lining across the fold of types I and II hose shall be not less than 400 percent, tested in accordance with 4.3.2.3, except that the low temperature stock in type III hose shall have an elongation of not less than 300 percent.

3.7.3 *Set.* The set of the lining shall not exceed 25 percent after being stretched to an elongation of 300 percent for type I and type II hose, and 250 percent for type III hose (see 4.3.2.4).

3.7.4 *Strength of lap.*

3.7.4.1 *Rubber lining.* The strength of the lap in the calendered rubber lining shall be not less than 75 pounds per inch of width (see 4.3.2.5).

3.7.4.2 *Cambric latex or rubber coated fabric.* The strength of the cambric latex or rubber coated lap shall be such that the break will occur in the fabric unless the stress is greater than 75 pounds. A parting of the lap by the shearing of the latex from the fabric without breaking the fabric at a value under 75 pounds shall constitute a failure. The test shall be made as described in 4.3.2.5.

3.7.5 *Accelerated aging.* The tensile strength and elongation of the rubber lining after having been aged as described in 4.3.2.2 shall be not less than 70 percent of the tensile strength and elongation before aging.

3.7.6 *Resistance to water, type III.*

3.7.6.1 *Swelling.* The lining of type III hose shall not increase in volume more than 5 percent or decrease in volume by more than 2 percent when immersed in distilled water as described in 4.3.5.1.

3.7.6.2 *Extraction.* The extractable material in type III hose shall not exceed 0.5 percent when boiled in distilled water as described in 4.3.5.2.

3.7.7 *Torsional stiffness and cold tension recovery, type III.* After low-temperature conditioning (94 hours at — 65° Fahrenheit (F)) the torsional stiffness ratio of the rubber lining shall not exceed 10, and the cold-tension recovery of the lining at — 65° F. shall be not less than 20 percent (see 4.3.4.2 through 4.3.4.2.3).

3.8 Low temperature flexibility, type III. All type III hose shall withstand bending at a temperature of — 65° F. without cracking or breaking (see 4.3.3).

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TABLE IV. *Requirements for hydrostatic tests applied to full length*

Size, inside diameter, inches:	Type I						Type II						Type III		
	Class A, double			Class B, single			Class A, double			Class B, single			Class B, single		
	1½	2½	3½	1½	2½	4	1½	2½	3½	1½	2½	4	1½	2½ and 3½	5 and 6
Test pressure for initial measurement, p.s.i.	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10
Final pressure for elongation expansion, twist, warp, and rise, p.s.i.	400	400	400	300	300	200	400	400	400	300	300	200	400	400	125
Test pressure, kink test, p.s.i.	350	350	250	300	300	200	350	350	250	300	300	200	350	350	75
Elongation, maximum, percent	8	8	8	10	10	10	8	8	8	10	10	10	10	10	10
Expansion, circumference, maximum, percent.	3	3	3	3	3	3	6	6	6	6	6	6	6	6	6
Twist[1] to right, maximum degrees per foot.	29	12½	12½	50	25	16	29	12½	12½	50	25	16	50	25	10
Twist[1] to left, maximum degrees per foot.	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Warp[2], maximum inches	20	20	20	20	20	20	20	20	20	20	20	20	20	20	4
Rise, maximum, inches	0	0	0	7	4	2	0	0	0	7	4	2	7	4	2

[1] A twist to the right, the direction that would tend to tighten the couplings, is indicated by a clockwise rotation of the coupling, looking from the hose toward the coupling. A twist to the left is indicated by a counterclockwise rotation.
A maximum twist to the left of 2° per foot will be permitted while pressure is being raised, but any final twist must be to the right.
[2] Deviation from a straight line.

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3.9 Resistance to fungus, types I and II. When specified (see 6.2), types I and II hose jackets, and cambric lining if used, shall be treated for resistance to fungus. The breaking strength of the cambric lining and types I and II hose jacket(s) as received shall not decrease more than 10 percent in either direction after leaching (see 4.3.6). The fungus treatment used shall have no detrimental effect on the physical properties of the jacket or lining.

3.10 Friction (adhesion). The adhesion between the backing and jacket, between the backing and lining, and between lining and jacket of types I, II, and III hose shall be not less than 12 pounds per 1.5 inch of width (see 4.3.2.6). This requirement is not intended to bar the construction of hose where there is no adhesion between the jacket and lining along the fold provided that the surface over which there is no adhesion is not greater than 35 percent of the total surface for class A hose. Class B hose shall have the tube fully attached around the entire surface.

3.11 Branding.

3.11.1 *Civil agencies.* Each length of hose shall be indelibly stenciled at the ends in letters 1 inch high with the manufacturer's name, the month and year of manufacture, and the words "Tested to _____lb/sq in." (The applicable hydrostatic pressure specified for its size in table IV as final pressure for elongation expansion, twist, warp, and rise psi). The lettering shall begin approximately 4 feet from the coupling. If a manufacturer produces hose at more than one factory, each length of hose shall have a distinctive marking, which may be in code, by which it may be identified as the product of a particular factory.

3.11.2 *Military agencies.* In addition to the branding required in 3.11.1, the identification marking shall be in accordance with MIL–STD–130.

3.12 Finish. The hose shall be finished in accordance with the manufacturer's standard practice. The finished hose shall be flexible and easily coiled.

3.13 Workmanship. The finished hose shall be free from defects which affect its appearance or may affect its serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 *Sampling.*

4.1.2 *Lot.* All hose lengths with couplings of the same size with the same construction materials, produced under essentially the same conditions, and offered for inspection at one time shall be considered a lot for purposes of inspection. For open end contracts, the size of the lot shall be constructed as the total number of those identical hose assemblies previously offered under the contract plus the number currently being offered for acceptance. The sample size for group B tests shall be based on the total constructed lot less the number of identical hose assemblies previously tested and accepted.

4.1.3 *Sampling for inspection.* The sample for inspection (see 4.1.2) shall be taken as described in Fed. Std. No. 601 except as modified by table V.

4.1.4 *Sampling for hydrostatic pressure tests of full lengths with couplings, group A*

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TABLE V. *Sampling for visual and dimensional inspection acceptable quality level (AQL) (approx) equals 2.5 percent defective for major defects, and 6.5 percent for major and minor defects combined*

Number of lengths of hose in lot	Number of lengths of hose in sample	AQL = 2.5 percent (defective units)		AQL = 6.5 percent (defective units) major & minor combined	
		Acceptance number	Rejection number major	Acceptance number	Rejection number
15 and under	10	0	1	1	2
16–40	15	1	2	2	3
41–110	25	1	2	3	4
111–300	35	2	3	5	6
301–800	50	3	4	6	7
801–3200	75	4	5	9	10

tests. Sample lengths shall be selected at random from the lot in accordance with table VI for tests in accordance with 4.3.1.1 through 4.3.1.5. One sample length out of each 15 or less sample lengths in each lot shall be subjected to the kink test specified in 4.3.1.6.

TABLE VI. *Sampling for group A tests*

Number of of lengths of hose in lot	Number of of lengths of hose in sample	AQL = 1.5 percent	
		Acceptance number	Rejection number
16–40	15	0	1
41–110	25	1	2
111–300	35	1	2
301–800	50	2	3
801–3200	75	3	4

4.1.5 *Sampling for group B tests.* Sample lengths for test shall be selected at random from the lot in accordance with table VII. Sample rubber test specimens of type III lining may be furnished from the manufacturer's assembled tubing run for tests in 4.3.4 through 4.3.4.2.3. The specimens shall have been manufactured from the same rubber and the same state of cure as the lining of the lot offered for inspection. Four 6-by 6-inch sheets are required with the same thickness as the lining of the hose being tested. A 2-foot length of type III hose jacket as woven, a 2-foot-length of uncoated cambric lining if used, and samples of the yarn used shall be furnished suitable for use in the tests of jacket construction (see 4.3.2).

TABLE VII. *Sampling for group B tests*

Number of of lengths of hose in lot	Number of of lengths of hose in sample	AQL = 4.0 percent	
		Acceptance number	Rejection number
5–40	5	0	1
41–110	7	0	1
111–300	10	0	1
301–800	15	1	2
801–3200	25	2	3

Sections and specimens shall be cut from each sample length of hose as follows:

(a) Two 3-foot sections for the tests in 4.3.2.1.

(b) One 3-foot section from which to cut specimens of the jacket, backing, and lining as applicable, for use in the tests in 4.3.2 through 4.3.2.6 and 4.3.5 through 4.3.5.2.

(c) One 2-foot section of type III hose for the test in 4.3.3, and one 2-foot section of type III hose from which to cut specimens for the tests in 4.3.4 through 4.3.4.2.3.

4.1.6 *Sampling for fungus resistance test.* An 8-foot section shall be taken from the manufacturer's assembled tubing run in each contract when fungus resistance (see 3.9) is specified, for use in the test in 4.3.6. The section shall be of the same material as the hose being furnished. The test shall be repeated if any change in material occurs.

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4.2 Inspection.

4.2.1 *Visual and dimensional inspection.* Each of the sample lengths, which do not involve tests selected in accordance with 4.1.3, shall be inspected to verify conformance to all of the requirements of this specification regarding both hose and couplings. The classification of defects applying to this particular hose, if available, and MIL–STD–166 shall be used to determine and evaluate defects through visual inspection. Any hose assembly containing one or more major visual or dimensional defects shall be rejected and counted against the AQL for major defective units.

4.2.1.1 *Fifty-foot length.* Any nominal 50-foot length of hose containing more than five minor defects shall cause the length to be rejected and counted against the AQL for total defective units. Any length containing five or less minor defects and no major defects shall be counted a minor defective against the AQL for total defective units but shall not be rejected.

4.2.1.2 *Lengths other than 50 feet.* For other lengths the number of minor defects which shall cause a length of hose to be rejected shall be proportional to the length. Computed fractional defects, but not less than 1, may be disregarded.

4.2.1.3 *Lot rejection.* If in any sample the number of defective lengths as defined above exceeds the acceptance number for either AQL for that sample, the lot represented by the sample shall be rejected. On open end contracts, hose assemblies accepted in previous lots shall not be rejected with the lot.

4.2.2 *Dimensions.*

4.2.2.1 *Length.* The distance between the inside edges of the couplings shall be determined as described in method 2411 of Fed. Std. No. 601 except that the measurement shall be made while a hydrostatic pressure of 10 psi is applied to the specimen as described in method 10311 of Fed. Std. No. 601.

4.2.2.2 *Inside diameter.* The id shall be determined as described in method 2341 of Fed. Std. No. 601 except that not less than two measurements shall be made on the unit or test unit. The measurements of hose with couplings shall be made when the 3-foot length is taken for the burst test, the 2-foot length for low temperature flexibility, or the 8-foot length for resistance to fungus.

4.3 Tests. Each of the samples selected in accordance with 4.1.4 through 4.1.6 shall be tested as specified in 4.3.1 through 4.3.5 as applicable, to determine conformance to this specification. Hoses bearing the label of the Underwriters' Laboratories, Inc. indicating compliance with the intent of UL Standard 19 for Cotton Rubber Lined Fire Hose need not be subjected to those tests which have been performed or confirmed by the UL label service.

4.3.1 *Group A, hydrostatic pressure tests.* The specimen shall be a full length of hose with couplings attached or the length remaining after removal of the portions for other tests, to which couplings have been attached by the manufacturer. The pressure applied to the specimen shall be as specified in table IV.

4.3.1.1 *Elongation.* The elongation of the hose shall be determined as described in method 10311 of Fed. Std. No. 601.

4.3.1.2 *Expansion.* The expansion of the hose shall be determined as described in method 10321 of Fed. Std. No. 601.

4.3.1.3 *Twist.* The twist of the hose shall be determined as described in method 10331 of Fed. Std. No. 601.

4.3.1.4 *Warp.* The warp of the hose shall be determined as described in method 10341 of Fed. Std. No. 601.

4.3.1.5 *Rise.* The rise of the hose shall be determined as described in method 10351 of Fed. Std. No. 601.

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4.3.1.6 *Kink.* A specimen which has been used in the tests described in 4.3.1.1 through 4.3.1.5, inclusive (see 4.1.4), shall be tested as described in method 10361 of Fed. Std. No. 601 to determine the strength of the hose when kinked. The pressure applied to the hose shall be as required in table IV.

4.3.2 *Group B, construction and physical property tests.* The thickness of the lining and backing shall be determined as described in method 2011 of Fed. Std. No. 601. The fabrics and yarns of the hose jackets (see 3.2.2, and 3.3.3 through 3.3.3.3), and the cambric (see 3.3.1.2), if used in the lining, shall be measured and tested in accordance with CCC–T–191 by the following methods for the following characteristics:

Method	*Characteristic*
5041	Weight (see 3.3.1.2).
5050.1	Construction (see 3.3.1.2).
5030	Thickness (see 3.3.1.2).
4050 and 4052	Twist (see 3.3.3.3.1 and 3.3.3.3.2).
4010	Length – weight relation for determination of denier (see 3.3.3.3.1) and counts of the single yarn (see 3.3.3.3.2).

After the removal of the specimens used in tests of the lining and backing as applicable, the remaining specimens of type III hose jacket, and the fabric used in cambric lining if used, shall be unraveled as necessary, and tested for consistency with the jacket and yarn samples furnished (see 4.1.5).

4.3.2.1 *Bursting strength.* One 3-foot section (see 4.1.5(a)) shall be tested at the pressure specified in table III, as described in method 10011, and one 3-foot section shall be tested at the same pressure, as described in method 10021, of Fed. Std. No. 601.

4.3.2.2 *Accelerated aging.* The resistance of the lining to accelerated aging shall be determined as described in method 7221 or method 7231 of Fed. Std. No. 601 except that the specimens shall be exposed for 166 ± 2 hours at 70° ± 1° centigrade (C) (158° ± 1.8° F) for types I and II hose, and 46 ± ¼ hours at 100° ± 2° C (212° ±3.6°F) for type III hose. The deterioration shall be measured by change in tensile strength, elongation, and hardness. Die I in method 4111 shall be used for cutting specimens (see 4.1.5(b)) for tensile strength and elongation, and the specimens shall be cut transversely across the fold so that the fold is approximately midway between the ends of the constructed portion of the specimen.

4.3.2.3 *Tensile strength and ultimate elongation.* The tensile strength and ultimate elongation of the lining shall be determined as described in methods 4111 and 4121 of Fed. Std. No. 601 respectively. For rubber lining die I in method 4111 shall be used for cutting the specimen; for the cambric latex or rubber coated lining, the coating on the passageway side (opposite side next to jacket) shall be carefully stripped from the fabric and die III in method 4111 shall be used. The specimen shall be cut transversely across the fold so that the fold is approximately midway between the ends of the constructed portion of the specimen.

4.3.2.4 *Set.* The set of the lining shall be determined as described in method 4411 of Fed. Std. No. 601. For rubber lining die I of method 4111 shall be used for cutting the specimen; for the cambric latex or rubber coated lining, the coating on the passageway side (opposite side next to jacket) shall be carefully stripped from the fabric and die III in method 4111 shall be used.

4.3.2.5 *Strength of lap.* The strength of the lap (splice) of the lining shall be determined as described in method 4311 of Fed. Std. No. 601. The cambric lining shall be a section including the fabric.

4.3.2.6 *Friction (adhesion).* The adhesion between the backing and jacket, between the backing and lining, and between the lin-

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ing and jacket shall be determined as described in method 8011 or method 8021 of Fed. Std. No. 601 except that (1) the middle of the specimen shall be 1.5 inches, (2) the specimen is prepared by marking with soft pencil two parallel lines 2½ inches apart following the filler yarns around the circumference of the hose. Cut the hose at the lap and along these lines so as to form a strip of a length equal to the circumference of the hose. The strip is laid out flat and the rubber lining cut through to the jacket along parallel lines 1.5 inches apart. The end of the central portion of the lining between the two parallel cuts is separated from the jacket for a short distance for gripping in the clamps.

4.3.3 *Low temperature flexibility, type III.* A 2-foot section of type III hose (see 4.1.5(c)) shall be maintained at a temperature of — 65° ± 2° F for a period of 24 hours. At the end of the 24 hour exposure period and while still maintained at the same exposure temperature, the hose shall be rapidly bent, 180° flat. The hose section shall then be examined for evidence of cracking or breaking.

4.3.4 *Physical properties after conditioning at — 65° F.* Six test specimens (see 4.1.5) shall be buffed to uniform thickness, 0.080 inch wide by 1½ inches long with a ¼-inch square stub at each end, and tested for torsional stiffness ratio and cold tension recovery.

4.3.4.1 *Torsional stiffness ratio.*

4.3.4.1.1 *Conditioning of specimens and test apparatus.* Specimens shall be conditioned for not less than 94 hours in a chamber maintained at — 65° ± 2° F. Test apparatus shall be conditioned for not less than 2 hours at — 65° ± 2° F.

4.3.4.1.2 *Test apparatus.* The test apparatus shall be a torsional tester conforming to Fed. Std. No. 601, method 5611 or 5612, except that only the torsion apparatus, the stand, and a suitable bracket to hold the lower clamp are required. Torsion wire of appropriate torsion constant shall be used.

4.3.4.1.3 *Test procedure.* The test procedure shall be as follows:

(a) With the specimen and the test apparatus at + 73° ± 5° F. the lower end of the specimen shall be secured in the rigidly mounted clamp of the test apparatus and the upper end shall be secured in the upper pinch clamp attached to the lower end of the torsional wire.

(b) The pointer shall be adjusted to the zero position by rotating the protractor scale. The torsional head shall be turned 180° and the pointer reading shall be recorded after a period of 10 seconds.

(c) After the above-mentioned test has been conducted, the specimen and test apparatus shall be conditioned (see 4.3.4.1.1). After conditioning is complete, the test shall be repeated on the conditioned specimen and tests shall be conducted at — 65° ± 0.5° F.

4.3.4.1.4 *Calculations.* The test shall be conducted on three specimens and the results averaged. If the torsional stiffness ratio for any one specimen is greater than 10, the ratio for each of the three specimens shall be within 20 percent of the average value of the three specimens. The torsional stiffness ratio at — 65° F. shall be calculated as follows:

$$\text{Torsional stiffness ratio at } -65^\circ \text{ F.} = \frac{\text{torsional stiffness factor at } -65^\circ \text{ F.}}{\text{torsional stiffness factor at } 73^\circ \text{ F.}}$$

where:

$$\text{torsional stiffness factor} = \frac{180^\circ \text{ angle of twist of specimen}}{\text{angle of twist of specimen}}$$

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4.3.4.2 *Cold tension recovery test.*

4.3.4.2.1 *Test apparatus.* The test instrument shall consist of a measuring board, similar to that illustrated in figure 1, on which several stretching devices can be mounted. Each stretching device shall have one movable clamp and one fixed clamp. The board shall have lines engraved at intervals corresponding to each 10 percent stretch, based on the length of the specimen between the ¼– by ¼– inch square stubs.

4.3.4.2.2 *Test procedure.* The test procedure shall be as follows:

(a) The measuring board shall be conditioned for not less than 2 hours at — 65° F. Three specimens, cut from the sample (see 4.1.5), at 73° to 78° F., shall be clamped in the stretching devices along the edges of the stubs, adjacent to the linear part, of the specimen and laid on the board. The movable clamp shall be pulled back until the 1½-inch long portion of the specimen has been stretched to 100 percent elongation and fixed in that position.

(b) The stretching devices and the specimens shall be conditioned in the low-temperature chamber for 7 days at — 65° F.

(c) With the test instrument and specimens still in the low-temperature chamber, the movable clamp shall be released from its fixed position, and the assembly shall be conditioned for an additional period of 30 minutes at — 65° F.

(d) The final length of the specimen shall be determined with the board, stretching devices, and specimens held at an angle of 15° from the vertical. The cold-tension recovery percentages for the three specimens shall be calculated and averaged. The average value shall be used to determine compliance with 3.7.7. If the cold tension recovery percentage for any one specimen is less than 20, the percentage for each of the three specimens shall be within 10 percent of the average for the three specimens. A cold tension recovery not within 10 percent of the average for the three specimens shall constitute failure of the test.

4.3.4.2.3 *Calculation.* The percentage of cold tension recovery shall be computed from the formula:

$$\text{Percentage cold tension recovery} = \frac{(Ls - Lf)}{(Ls - Lo)} \times 100$$

where:

Ls = stretched length of specimen.
Lf = final length of specimen.
Lo = initial length of specimen.

4.3.5 *Resistance to water, type III.* Specimens shall be taken from hose sections (see 4.1.5(b)) and tested for swelling and water extraction (solvent action).

4.3.5.1 *Swelling.* The increase in volume of the lining shall be determined as described in method 6211 of Fed. Std. No. 601. The immersion medium shall be medium 10 of method 6001 of Fed. Std. No. 601. The latex or rubber coat shall be stripped from the fabric in performing this test.

4.3.5.2 *Water extraction.* The solvent action of water on the lining shall be determined as described in method 6621 of Fed. Std. No. 601. The immersion method shall be media 10 of method 6001 of Fed. Std. No. 601. The latex or rubber coat shall be stripped from the fabric in performing this test.

4.3.6 *Fungus resistance, types I and II.* The jacket material shall be tested for resistance to fungus as described in method 5762 of CCC–T–191 using untreated and uncoated samples of hose jacket, and cambric lining

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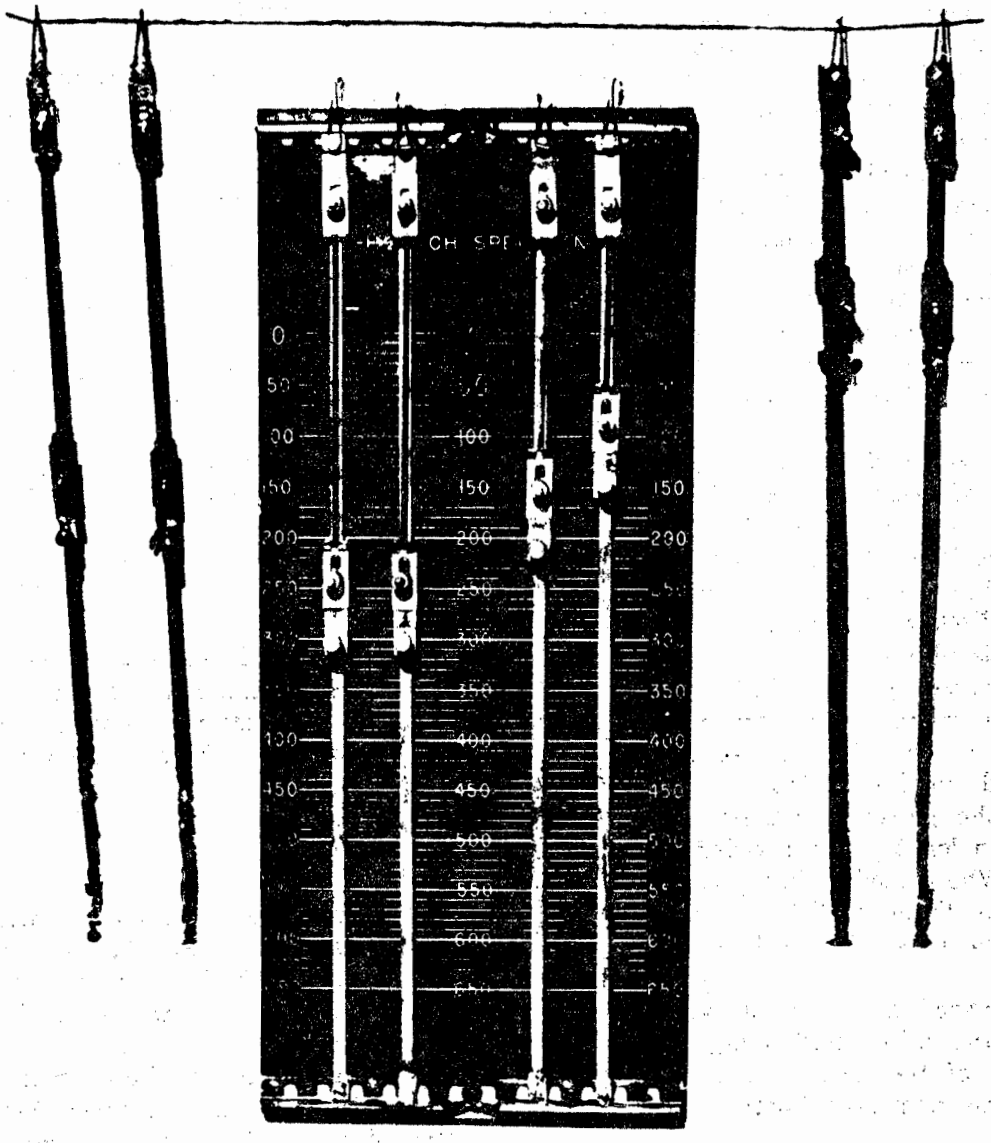


Figure 1. Tension recovery apparatus.

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if used, as loomed in lieu of 12– to 14-ounce cotton duck, as a reference standard. The incubation period shall be 2, 4, and 6 weeks, for 3, 3, and 4 specimens each, respectively. The procedure shall be as follows:

From the sample length of hose furnished, prepare two sets of ten specimens each; one set for breaking strength controls and one set for leaching, inoculation, and incubation. The lining shall be removed from the break strength control samples before cutting to size. The removal of the lining must be done in a manner that the jacket(s) will not be injured, and on the cambric lining so that the fabric is not injured. Cut the specimens 1½ inches wide, and 6 inches long, and then ravel the specimen to 1 inch width. The cambric lining may be cut to 1 inch width and not ravelled. Control breaking strength will then be made on the jackets and on the cambric coated lining where used. The second set of specimens shall be cut not less than 1½ inches wide by 8 inches long. The lining shall be left intact on these specimens during the leaching, inoculation, and incubation. Following incubation, washing, and drying, the lining shall be carefully removed so as not to injure the jackets nor the fabric in the cambric lining where used, and to provide at least 6 inches of length of undamaged specimen. After removal of the lining, the jacket specimens shall be cut 6 inches in length, and ravelled to 1 inch in width. Where cambric lining is used, the specimens of lining shall be cut to 6 inches in length by 1 inch in width.

4.4 Inspection of preparation for delivery. The preservation, packaging, packing, and marking of the hose shall be inspected to determine compliance with section 5.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the levels of preservation, packaging, and packing used are defined in Fed. Std. No. 102.)

5.1 Preservation and packaging. The preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.* The hose shall be preserved and packaged in accordance with level A of MIL–H–775. When hose is subjected to hydrostatic tests at the manufacturer's plant, the hose shall be carefully drained and the jackets thoroughly dried before preparation for shipment.

5.1.2 *Level C.* The hose shall be prepared for shipment to afford adequate protection against damage from the supply source to the first receiving activity. When hose is subjected to hydrostatic tests at the manufacturer's plant, the hose shall be carefully drained and the jackets thoroughly dried before preparation for shipment.

5.2 Packing. The packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* The hose shall be packed in accordance with level A of MIL–H–775.

5.2.2 *Level B.* The hose shall be packed in accordance with level B of MIL–H–775.

5.2.3 *Level C.* The hose shall be packed or prepared for shipment in a manner to insure arrival at destination in satisfactory condition, and which will be acceptable to carriers at lowest rates.

5.3 Marking.

5.3.1 *Nonmilitary agencies.* Unless otherwise specified (see 6.2), shipping containers shall be marked with the name, type, class, and size (diameter) of the commodity as defined by the contract or order under which the shipment is made, the quantity of each type, class, and size contained therein, the name of the contractor, the number of the contract order, the name and address of the consignee, and the gross weight.

5.3.2 *Military agencies.* Interior packages and shipping containers shall be marked in accordance with MIL–STD–129.

6. NOTES

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6.1 Intended use. Class A hose is intended for use on pumping engines and in places where service conditions require the additional protection against wear by the extra jacket. Types I and II, class B hose is intended for use at fire hydrants, stand-pipes, and places where service conditions are not considered severe. Type III hose is intended for general use and for use on fire fighting equipment.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type, class, size, and length of hose required (see 1.2.1 and 3.3.5).

(*c*) When cambric fabric coated lining may be used (see 3.3.1).

(*d*) When couplings for type III, sizes 1½ and 2½, shall be type D1 (see 3.4).

(*e*) When couplings are other than as specified and whether both couplings shall be male or female (see 3.4).

(*f*) When coupling thread form for shipboard use is required (see 3.4).

(*g*) When fungus resistance is required for types I and II hose and for cambric lining (see 3.9).

(*h*) Level of preservation and packaging, and level of packing required (see 5.1 and 5.2).

(*i*) When shipping container shall be marked other than specified in 5.3.1.

6.3 Transportation description. The transportation description applicable to this item is:

Rail:

Hose, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.

Motor:

Hose, not otherwise indexed.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

MILITARY CUSTODIANS:

Army—MO (ERDL)

Navy—Docks —

Air Force—ASD

Review interests:

Army—MO
Navy—YD, SH
Air Force—11
DSA—CS

User interests:

Army—MI
Navy—MC, CG

Review/user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DOD Standardization Documents.

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